

Reconnaissance & Vulnerability Scanning Report

Internship Task Submission

Date: 04 June 2026

1. Objective

The objective of this task was to perform active reconnaissance and vulnerability scanning on safe test systems using cybersecurity tools such as Nmap and Nikto. The assessment focused on identifying open ports, running services, server configurations, and potential security vulnerabilities.

2. Tools Used

Tool	Purpose
Nmap	Port scanning, service detection, OS detection, and enumeration
Nikto	Web server vulnerability scanning and security misconfiguration detection
Kali Linux	Operating system used for performing security assessment tasks

3. Nmap Scan Results

An Nmap scan was performed against scanme.nmap.org using the command:
nmap -sC -sV -O scanme.nmap.org

```
(abhisith@kali) ~$ nmap -sC -sV -O scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-03 21:06 IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|   256 33:fa:91:0f:e0:a1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-favicon: Nmap Project
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Device type: general purpose
Running (JUST GUESSING): Linux 5.X (85%)
OS CPE: cpe:/o:linux:linux_kernel:5
Aggressive OS guesses: Linux 5.0 - 5.14 (85%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.91 seconds
```

Figure 1: Nmap Port and Service Enumeration Results

Key Findings from Nmap Scan

- Port 22/tcp was open running OpenSSH 6.6.1p1 service.
- Port 80/tcp was open running Apache HTTP Server 2.4.7 on Ubuntu Linux.
- Filtered ports such as SMTP, NetBIOS, and Microsoft-DS were identified.

- OS detection suggested the target system was running Linux Kernel 5.x.
- Service enumeration successfully identified versions and banners.

4. Nikto Vulnerability Scan Results

A Nikto web vulnerability scan was performed against the test web application using the command:

nikto -h http://testaspnet.vulnweb.com

```
(abhisith@kali)-[~]
$ nikto -h http://testaspnet.vulnweb.com
- Nikto v2.5.0

+ Multiple IPs found: 44.238.29.244, 64:ff9b::2cee:1df4
+ Target IP: 44.238.29.244
+ Target Hostname: testaspnet.vulnweb.com
+ Target Port: 80
+ Start Time: 2026-06-03 21:18:47 (GMT5.5)

+ Server: Microsoft-IIS/8.5
+ /: Retrieved x-aspnet-version header: 2.0.50727.
+ /: Retrieved x-powered-by header: ASP.NET.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type.
+ See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ RFC-1918 /aspnet_client: IP address found in the 'location' header. The IP is "10.0.0.14". See: https://portswigger.net/kb/issues/00600300_private-ip-addresses-disclosed
+ /aspnet_client: The web server may reveal its internal or real IP in the Location header via a request to with HTTP/1.0. The value is "10.0.0.14". See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2000-0649
+ OPTIONS: Allowed HTTP Methods: OPTIONS, TRACE, GET, HEAD, POST .
+ OPTIONS: Public HTTP Methods: OPTIONS, TRACE, GET, HEAD, POST .
+ /test.txt: This might be interesting.
+ ERROR: Error limit (20) reached for host, giving up. Last error: opening stream: can't connect (connect error): Network is unreachable
+ Scan terminated: 12 error(s) and 9 item(s) reported on remote host
+ End Time: 2026-06-03 21:45:22 (GMT5.5) (1595 seconds)

+ 1 host(s) tested
```

Figure 2: Nikto Vulnerability Scan Results

Key Findings from Nikto Scan

- The server was identified as Microsoft-IIS/8.5.
- The X-Frame-Options header was missing, increasing clickjacking risk.
- The X-Content-Type-Options header was not configured properly.
- HTTP TRACE method was enabled, which may expose the server to Cross-Site Tracing attacks.
- Internal IP address disclosure was detected in HTTP headers.
- Potentially interesting file /test.txt was discovered.

5. Security Risks Identified

Risk	Description
Missing Security Headers	Could allow attacks such as clickjacking and MIME-type sniffing.
TRACE Method Enabled	Can be abused for Cross-Site Tracing (XST) attacks.
Information Disclosure	Internal IP leakage may help attackers gather network information.
Open Services	Publicly exposed services increase the attack surface.

6. Recommendations

- Disable unnecessary open ports and services.

- Implement security headers such as X-Frame-Options and X-Content-Type-Options.
- Disable HTTP TRACE method on the web server.
- Avoid exposing internal IP addresses in server responses.
- Regularly perform vulnerability scanning and patch management.

7. Conclusion

The reconnaissance and vulnerability scanning task successfully identified open ports, running services, server details, and several security misconfigurations. The activity demonstrated practical usage of Nmap and Nikto for cybersecurity assessment and vulnerability analysis in a safe testing environment.